Online Experiments for Language Scientists

Lecture 8: Priming and overspecification

Jenny Culbertson

jennifer.culbertson@ed.ac.uk

Questions about UG assessment 1?

Ask now!

Office hours for OELS questions:

- Tuesday 4th : 11-1pm, ~~just drop in (1.14 DSB)~~ OELS office hours | Meeting-Join | Microsoft Teams

- Last chance for email questions:
  - 4pm on Monday 3rd November!

Loy & Smith (2021)

Jia Loy
(now works in industry)

Loy, J. E., & Smith, K. (2021). Speakers Align With Their Partner's Overspecification During Interaction. *Cognitive Science*, 45, e13065.

- Design:
  - Director-matcher game: participant with partner
  - Confederate priming experiments
- Research question: Do people copy their partner's tendency to **overspecify**?

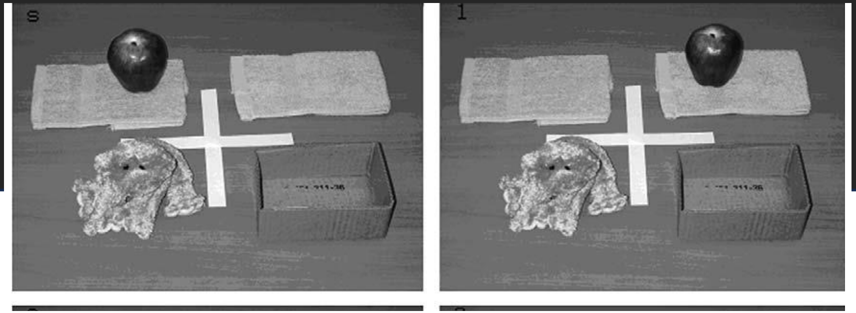
Overspecification

Overspecification: providing more information than what is strictly needed.

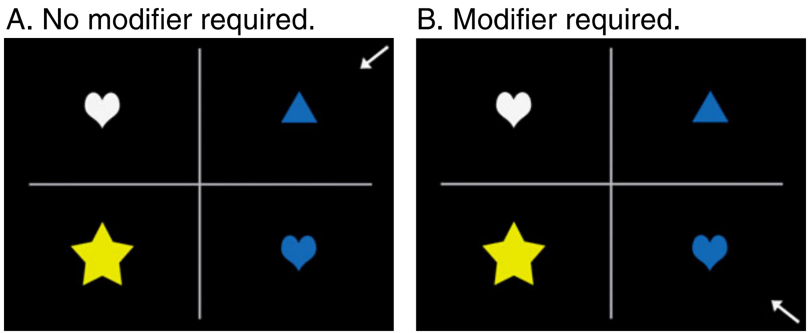
e.g., Engelhardt, Bailey & Ferreira (2006) Do speakers and listeners observe the Gricean Maxim of Quantity? *Journal of Memory and Language, 54, 554-573.*

(also Engelhardt & Ferreira 2014)

It's **<u>not</u>** very efficient!



"Put the apple in the box" vs.
"Put the apple that's on the towel in the box"



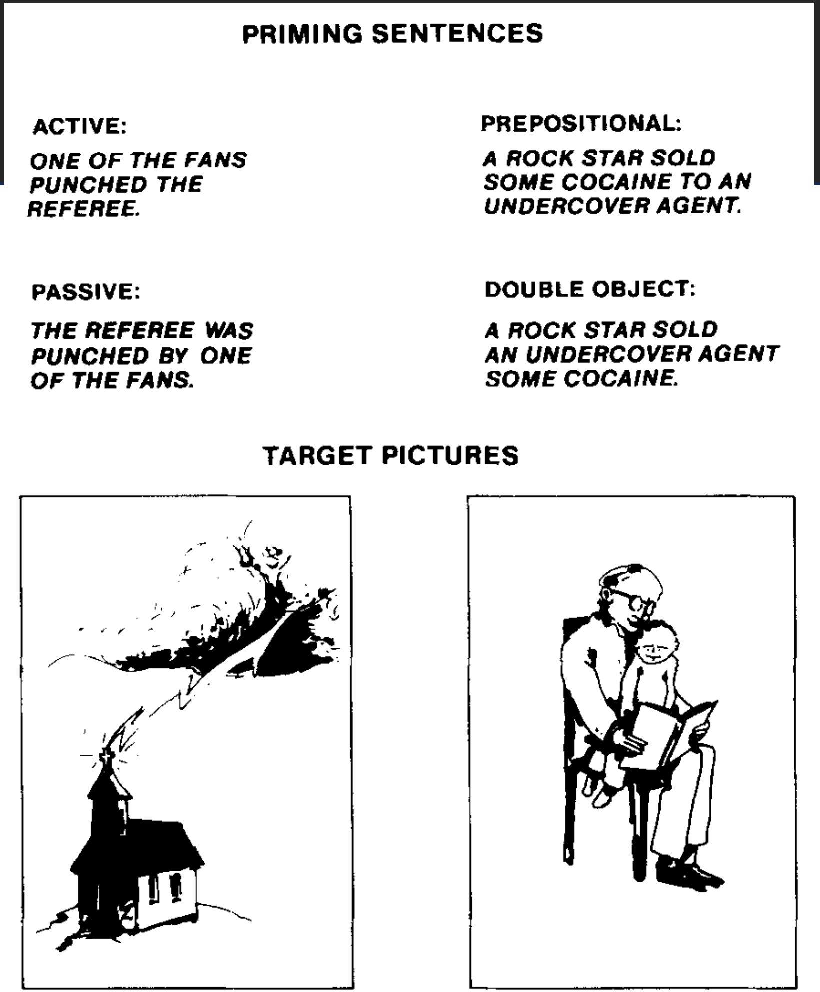
"The (blue) triangle" vs.
"The (small) blue heart"

Priming

Priming: people repeat what they have recently heard or produced

Structural priming: people repeat abstract structures they have recently heard or produced

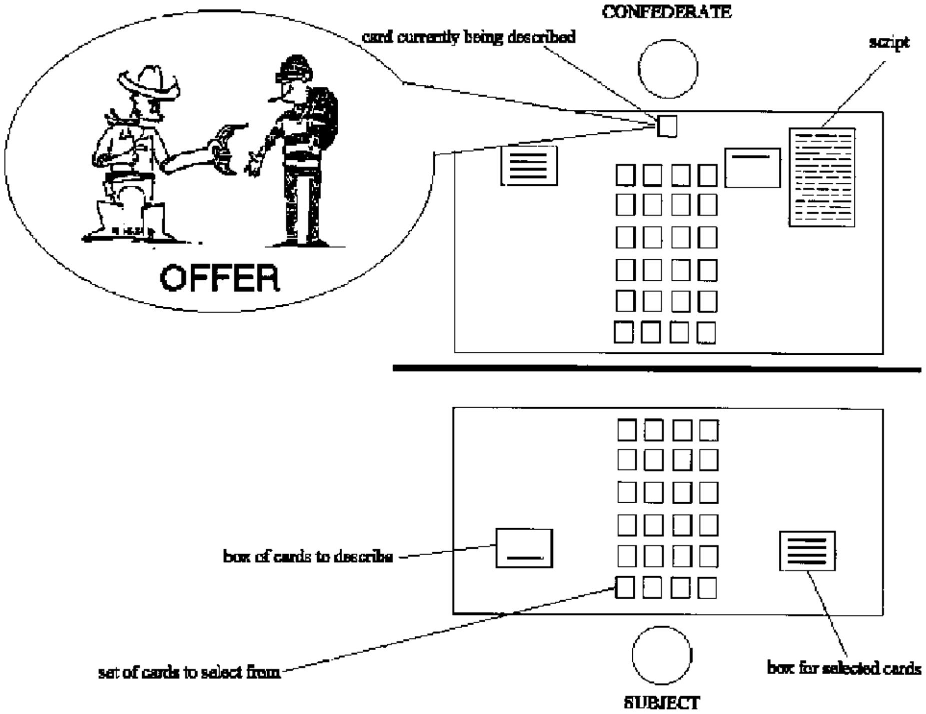
e.g. Bock, J. K. (1986). Syntactic persistence in language production. *Cognitive Psychology, 18,* 355-387.

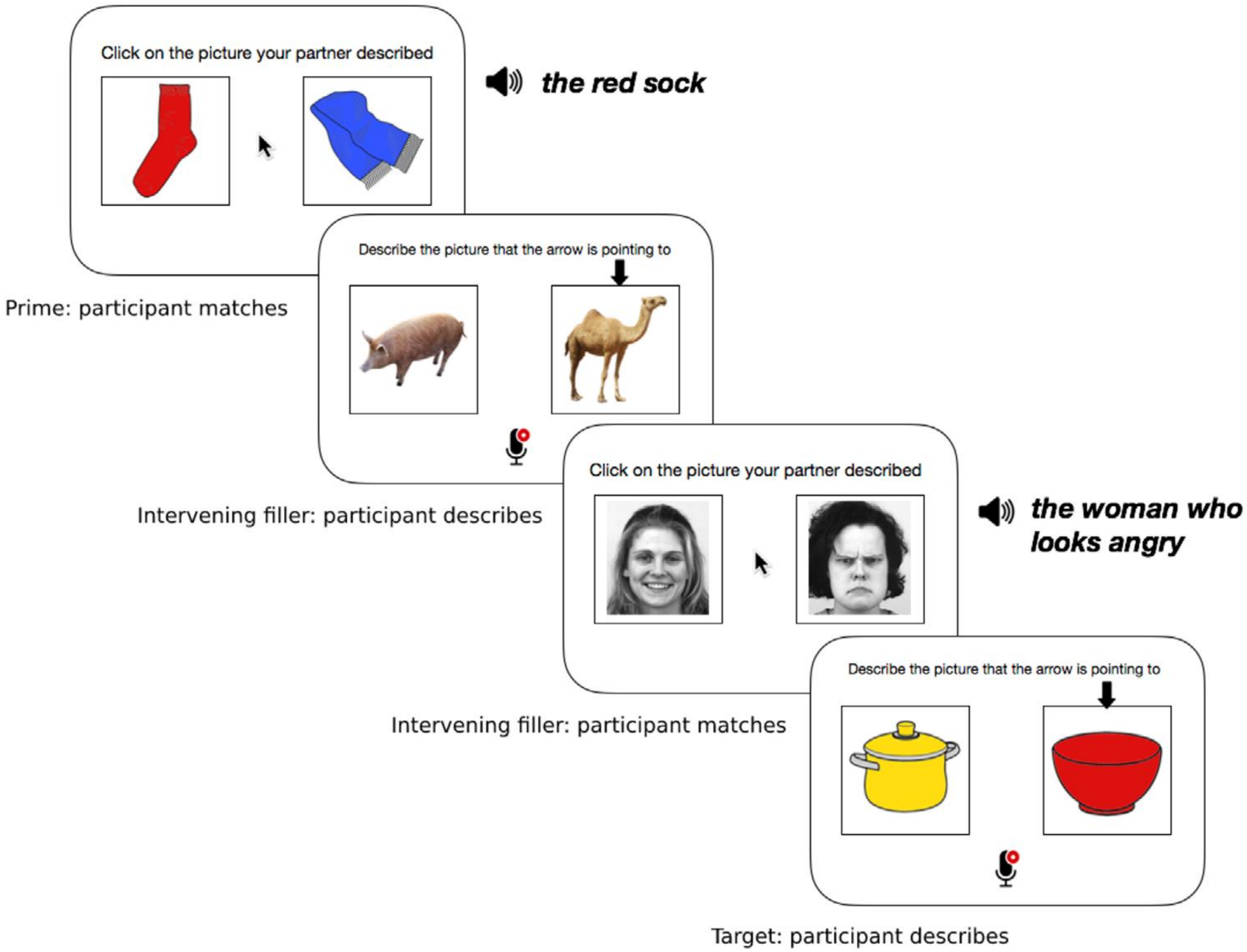


Confederate priming

Confederate: "a person one works with, especially in something secret or illegal; an accomplice"

Branigan, H. P., Pickering, M. J., & Cleland, A. A. (2000). Syntactic coordination in dialogue. *Cognition, 75,* B13-25.




Click on the picture your partner described
the red sock
Prime: participant matches
Describe the picture that the arrow is pointing to
Intervening filler: participant describes
Click on the picture your partner described
the woman who looks angry
Intervening filler: participant matches
Describe the picture that the arrow is pointing to
Target: participant describes

Demo using this week’s code

Loy & Smith (2021)

Manipulation: partner’s tendency to overspecify

- **Exps 1, 2**: colour, partner either consistently overspecifies (uses colour adjectives) or not (uses bare nouns)
- **Exp 3**: size, partner either consistently overspecifies (uses size adjectives) or not (uses bare nouns)
- **Exp 4, 5**: colour, partner switches behaviour mid-way through experiment

Loy & Smith (2021): details

Exp 1: *lab-based*

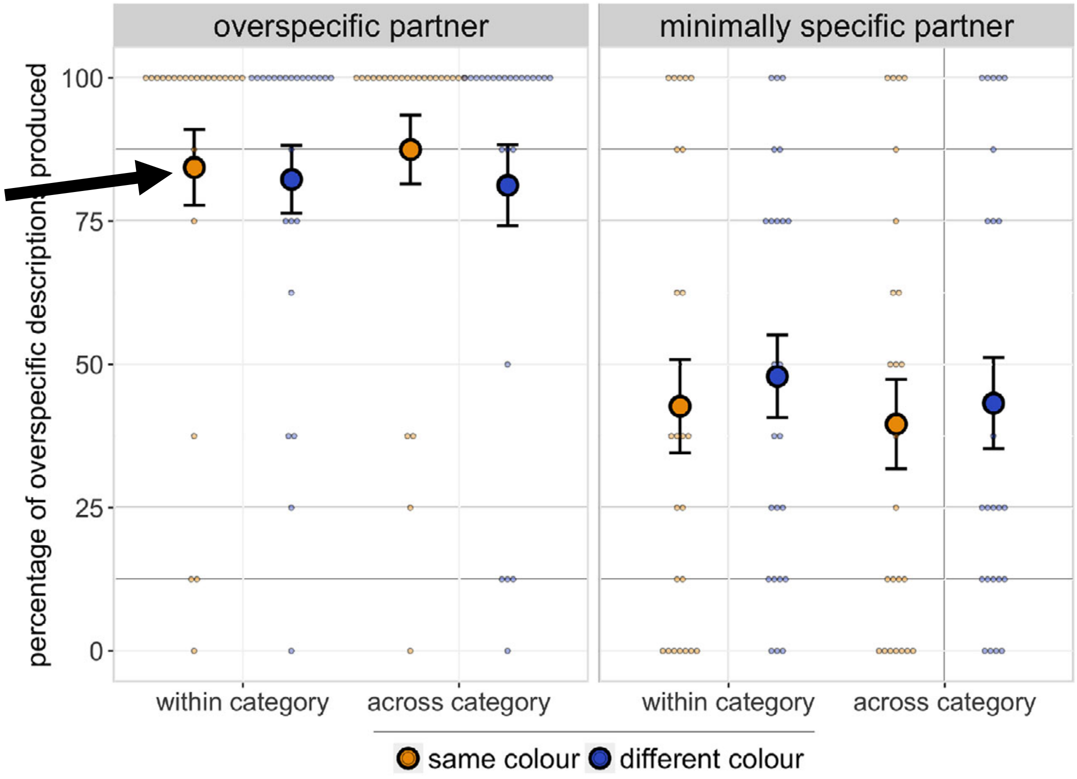
- N=24 per condition after exclusions
- Paid £6

Exps 2-5: *MTurk*

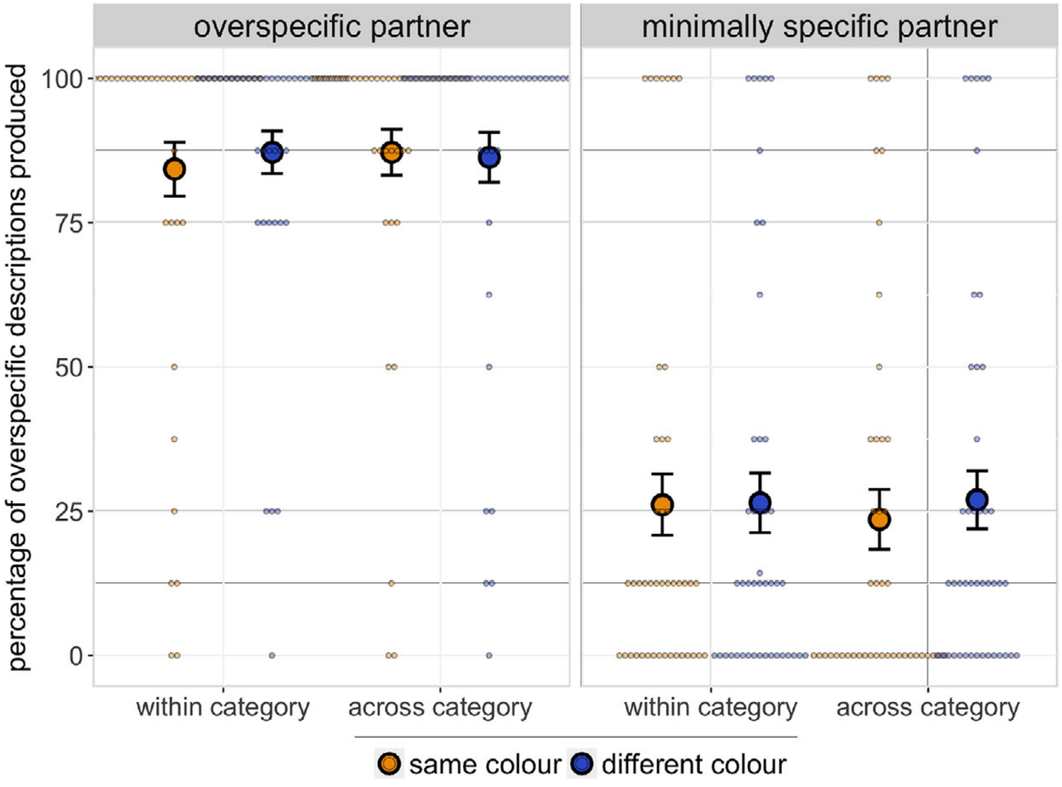
- N≈50 per condition after exclusions
- Paid $6

Experiment 1: lab, colour

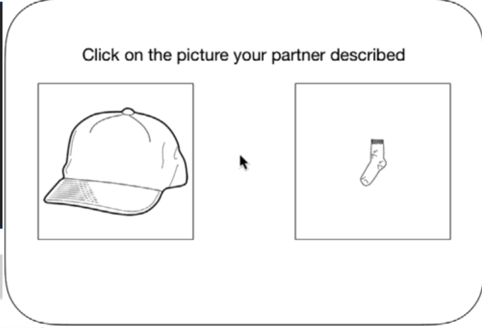
Same colour
Same category
…might expect
more priming



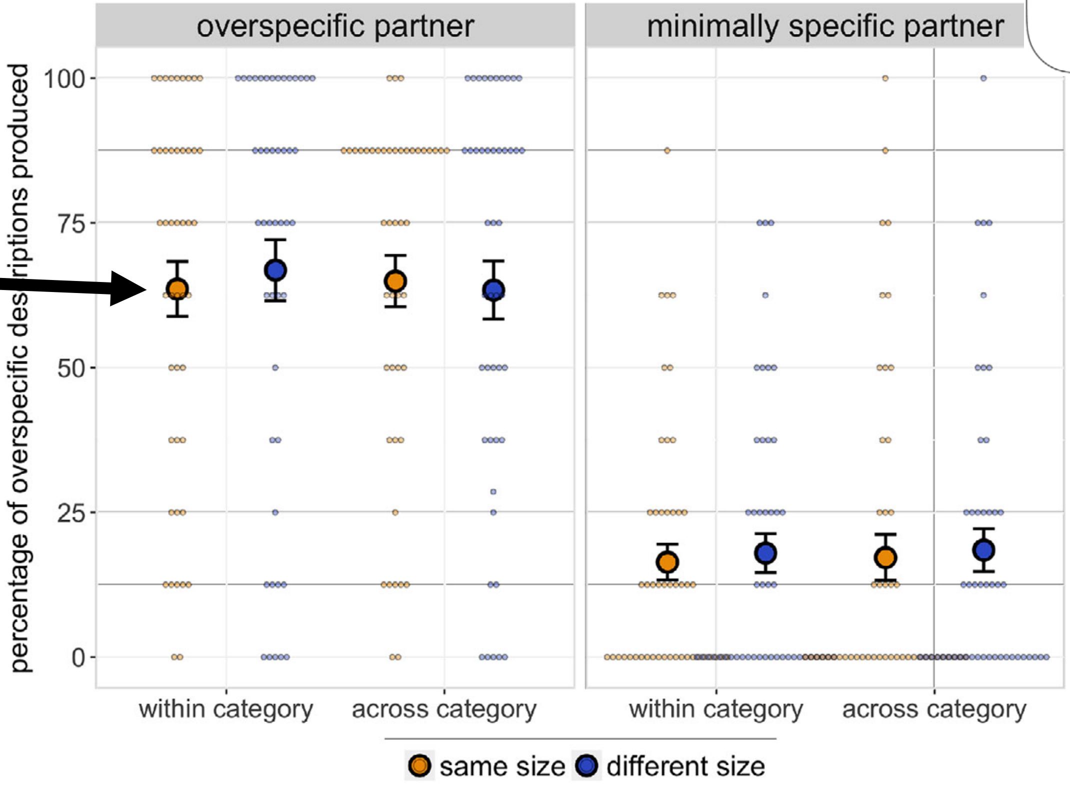
Experiment 2: online, colour

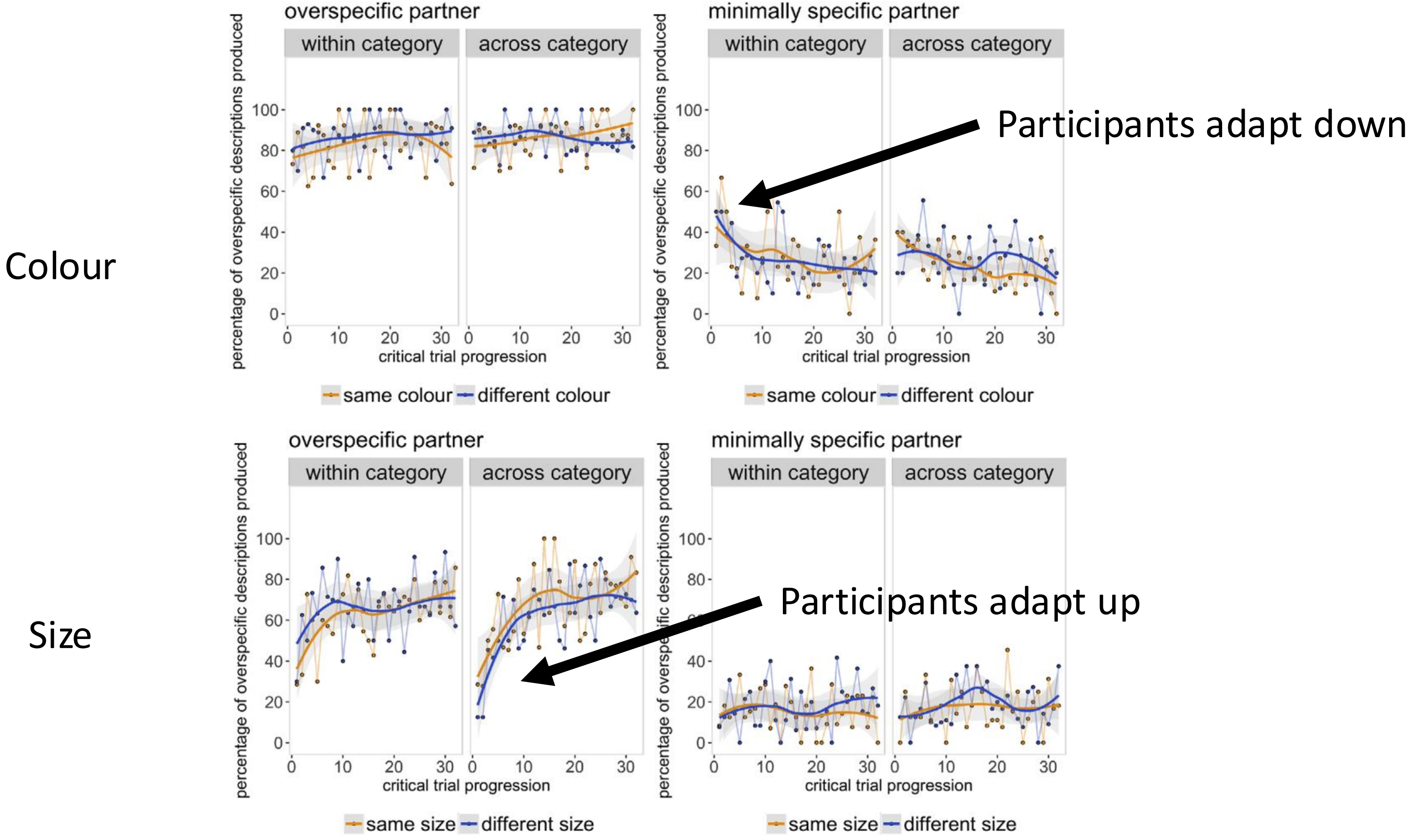


Experiment 3: online, size

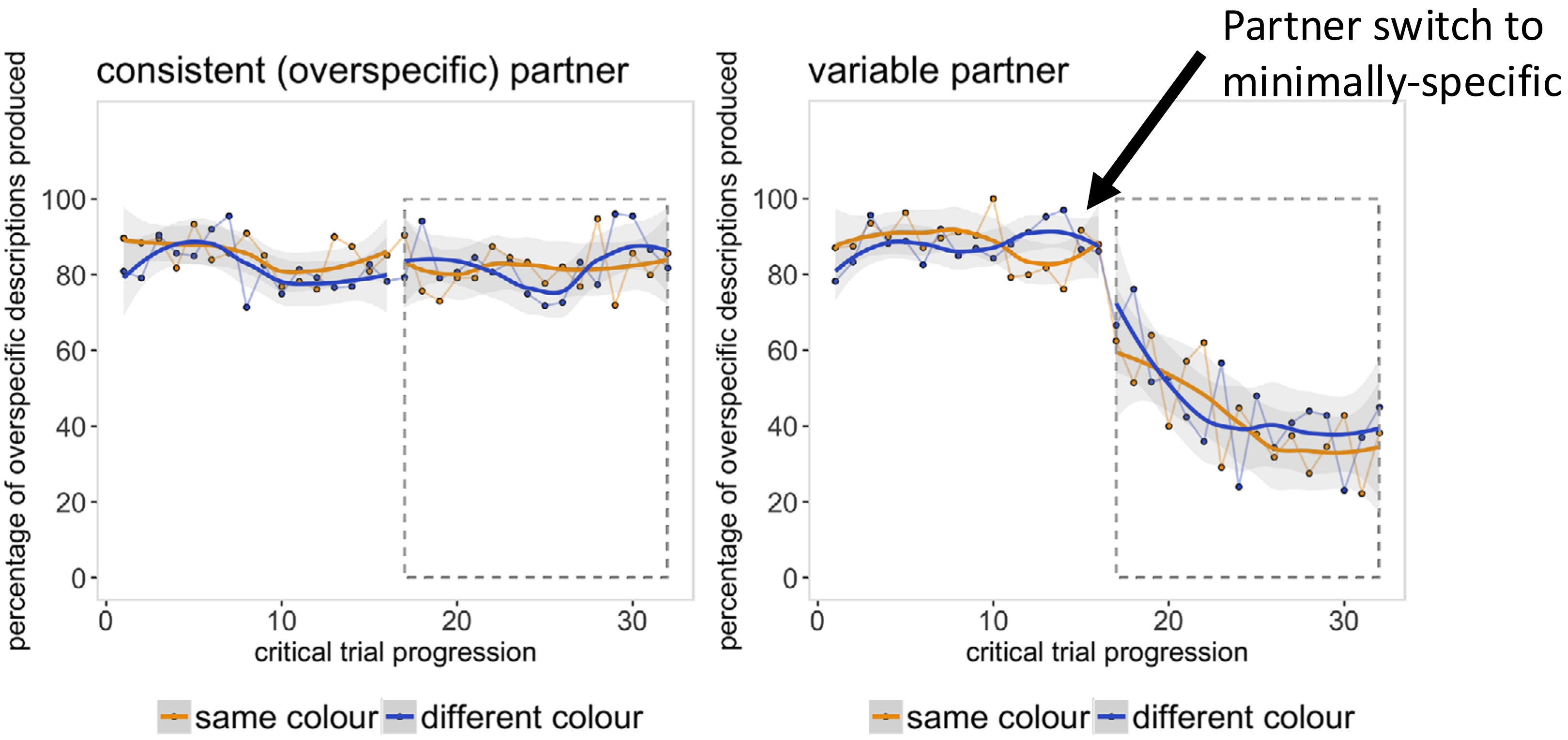


Less overspecification at the start

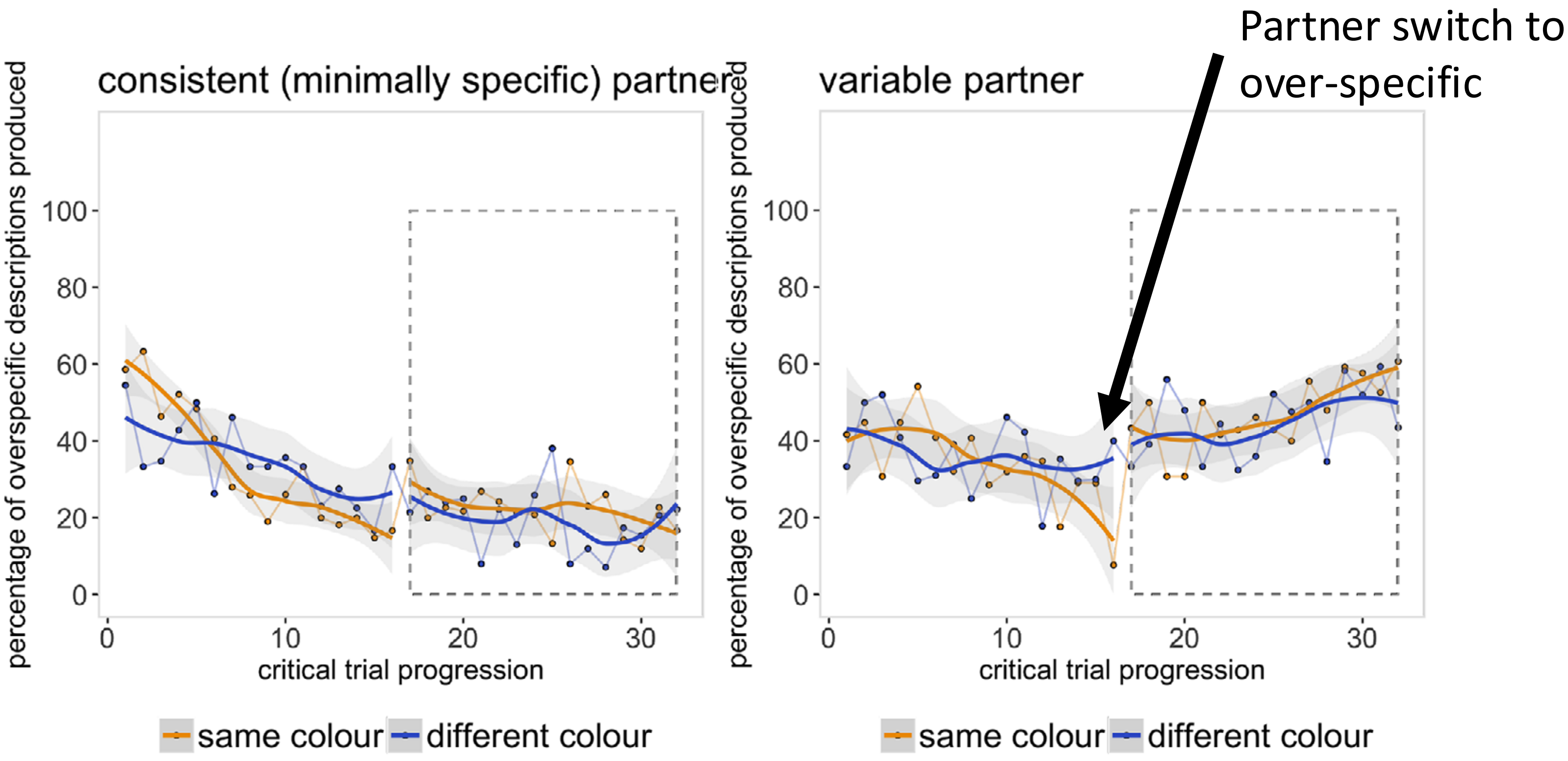



Colour
Size
overspecific partner
minimally specific partner
within category
across category
percentage of overspecific descriptions produced
critical trial progression
same colour
different colour
same size
different size
Participants adapt down
Participants adapt up

Experiment 4: switch to minimally-specific



Experiment 5: switch to over-specific



Loy & Smith's conclusions

1. Overspecifying (or not) can be **<u>primed</u>** during interaction
    - Including if their partner switches behaviour mid-way through the experiment

2. **<u>Social effects</u>** are a *constraint* on people's tendency to behave in an optimally efficient manner in communication

Time for Q&A/discussion on this week’s reading

Next up

Lab

- A confederate priming experiment, recording spoken responses

Next week

- Language evolution by iterated learning